

TEND ROBOTICS

Keeping elderly people safe in their own homes.

Pre-read for Raj

Tend keeps elderly people safe in their own homes by wiring ambient risk detection to a live clinician who responds within seconds, on a path toward one remote nurse caring for 100 patients.

Summary

- Tend is the platform that lets elder care nurses see 10x the patients in their own homes.
- Demand for elder care is insatiable and growing. Current systems are broken with insufficient labour and an unsustainable cost of care. The system is crying out for better tools.
- I would use the time and funds to deeply understand the problem, talk to at least 300 carers, providers and families, and make a call on whether this is the problem I want to commit to for the next decade.

The framework I'd apply during Frontier

I've been applying a [framework](#) from Hardy Michel (Marloo) while looking for a problem to solve recently, asking the following questions to validate whether I'm working on the right problem:

- **Is there a real, painful, high-frequency problem to solve?** Not one I've convinced myself exists, but one that pulls me in.
- **Do I have unique insights that compound?** Knowledge or perspective that gives me an unfair advantage.
- **Is this interesting enough to be my life's work?** Am I passionate enough to spend the next decade on this?

This framework has led me to this problem, a heartbreaking and globally painful problem that's close to my heart and that I'm uniquely placed to solve with my physical AI & mechatronics engineering background. **I'd use the three months at Frontier to make certain that my assessment is correct, and dive deep into the industry, understanding it deeply enough to make a generation-defining impact. That will look like:**

Months 1 & 2: Understand deeply

- Running deep discovery with carers, providers, and families to pin down the exact moments where a live clinician through a device creates value, and the ones it does not. Targeting 300 conversations over the 3 months.
- Build the network. Convert existing relationships across government health agencies, home-care providers, and the healthcare community into pilot and evidence partners.

- Commit fully. Move to Tend full-time, with the IP held cleanly in the company and my full attention on the build.

Month 3: Testing demand

- Build a scrappy MVP and get at least 5 firms on a paid pilot, to validate if there is real pull.

The problem

Somewhere tonight, an older person is on the floor of their own home, and no one knows yet. That gap, the hours between visits when nobody is watching, is the problem Tend exists to close.

Older people want to stay home, and the system cannot keep them there safely. This wish is near-universal in the West.

- In the US, nearly nine in ten seniors want to age in place rather than enter a facility.²

The workforce to honour that wish does not exist. The labour shortage is a systemic gap that needs new tools.

- In the United States, over 59 percent of home care agencies report ongoing caregiver shortages.² The average agency is turning away 510 hours a month of work they cannot staff.¹⁸
- Australia will be short 110,000 direct-care workers by 2036 and more than 400,000 by 2050.
- In New Zealand, aged residential care holds the worst nursing vacancy rate in the health system, 15 percent in 2023.³

The cost of care is crushing, and many families pick up the slack.

- Residential care runs ~NZD60-90k in New Zealand,⁵ ~AUD70k-100k in Australia²⁰ and ~USD115-130k in the US.¹⁹
- Families face a choice between two hard options: drawing down on the family estate for elder care, or supporting their loved one entirely on their own. AARP estimated that unpaid family caregivers accounted for 49.5 billion hours globally in 2024, at an economic value of >\$1 trillion.

And the quiet cost is loneliness.

- Isolation raises dementia risk by about 50 percent, heart disease by 29 percent, and stroke by 32 percent.⁶

Physical AI can solve this

Physical AI and robotics shows promise in the in-home care space; I've covered the key players in full in the appendix. The gap in the Western markets (AU, USA, Europe, NZ) is remote patient monitoring & real-time escalation of issues of concern.

- **Competitors stop at detection and inference:** Companionship, medication reminders, and passive alerts are well-covered ground. However, nobody in this space is capturing identifying potential health risks in real time (there's an opportunity to do this

with newly FDA-approved radar devices) and then escalating to a carer in seconds to attend to the issue.

- **Tend will identify risks in real time and escalate to a clinician to raise the alarm:** Remote monitoring with outsourced labour is possible, an offshore nurse could act as a triage and escalation specialist, not a care worker.
- **Orchestrating** across a smart home. We're at the very beginning of the Physical AI wave; It's reasonable to expect household humanoids being deployed at scale in the next five years, and IoT devices are already commonplace. Tend will act as the intelligence layer that makes embodied AI deployable in a home care setting.
- **The moat is regulatory.** It is the vetted clinical network, clinical governance, liability cover, and payer contracts that turn detection into a live response.

The vision

Phase 1: sensing plus companion plus a live clinician

We orchestrate proven components rather than build them. The existing carers keep visiting but now only when required. Our system watches the gaps, and can alert a clinician live.

Phase 1 capabilities:

- Detect falls, including someone who has fallen or is found on the ground between carer visits.
- Prompt medication at the right time and confirm it was taken, where memory is unreliable.
- Check that the person is eating and moving through their day.
- Track breathing rate, heart rate, and movement through contactless radar. No wearable, nothing touching the person.
- Watch daily patterns like room transitions, bathroom use, and time out of bed, and flag changes.
- Flag risk signals and raise the alarm to a live clinician.

Phase 2: mobility and one nurse for one hundred homes

Once families trust a Tend device in the home, we add mobility and non-contact physical tasks.

We move into embodied robotics only once value is proven and the brand is trusted. We wait for bodies like Neo to be ready.

These capabilities stay non-contact by design for now, which keeps us clear of medical-device regulation. (see Appendix: Regulatory position)

Phase 2 capabilities:

- Move to the person and the incident. Bathrooms are a major risk area.
- Carry the nurse to the person for telepresence through teleoperation.
- Guide and lead: kitchen to a meal, door to a visitor, bed at night.

- Fetch and carry non-clinical objects around the home.
- Visually inspect the environment, then prompt and confirm at close range.

The destination is one remote nurse caring for 100 patients.

As the technology keeps advancing, a robot will one day shower a person and cook them a meal. The tech is almost there. People are not ready yet, but trust is built slowly, and Tend will be the one they trust to do it.

Teleoperation is the bridge to autonomy. Early Phase 2 tasks run with a human in the loop, exactly as Neo ships today. When an incident happens, a nurse can take control through teleoperation and be present in seconds. As models improve, more of each task runs on its own, and one operator supervises more homes. The nurse-to-homes ratio that drives margin is a function of PAI advancement.

Appendix 1: Competitive landscape

The market splits into companion robots, monitoring software, and home humanoids. None closes the loop Tend closes.

Intuition Robotics (ElliQ). A tabletop companion offering reminders, fall-risk assessment, and a check-engine-light alert to clinicians. It has raised about USD 85 million.¹⁴

Washington authorised a statewide Medicaid code for ElliQ, the first social AI robot fully reimbursed by Medicaid.¹⁵ We orchestrate this device in Phase 1 if it stays the strongest option.

ElliQ is expanding internationally. In September 2025, Intuition Robotics partnered with the trading house Kanematsu to bring a localised ElliQ to Japan, the world's oldest population, with a 2026 launch.¹⁶

It is not mobile and does not support teleoperation. It flags a human; it does not connect one who responds live. The device we will orchestrate is proving itself in the exact markets we care about.

1X (Neo) is the leading in-home humanoid. It ships in 2026, teleoperated first, with elderly assistance named as a target use case. It brings the body, not the care loop.⁷

We leverage this hardware if it stays ahead, and switch to another maker if it does not. Staying body-agnostic keeps that choice ours.

Zuowei (Shenzhen Zuowei Technology) shows where China is going. It builds smart toileting and bathing robots and a broader eldercare platform, claiming China's first systematic range across six core needs: toileting, bathing, eating, transfers, walking, and dressing.

This is the contact-care frontier we deliberately avoid for now. It marks where the market heads once trust and regulation catch up.

Andromeda Robotics (Abi) is a companion, and a possible partner. It places friendly companion robots in aged care, with no monitoring or teleoperation and heavy EU biometric exposure. We have a relationship with Andromeda and may partner to use their operating system if needed.

Sensi.AI monitors remotely, but audio only. It detects and alerts. It does not connect to a clinician who intervenes.

CareClever (Cutii) is the cautionary tale. A capable mobile companion sold B2C at 90 euro a month. Good product, wrong model. It died on consumer subscription and detection liability.

The gap on one line: companionship and monitoring are table stakes. None wire detection to a live professional who responds in seconds. That position is open.

Regulatory position

The claim this rests on

The pre-read states that Tend is not a medical device. That claim is load-bearing, so it needs to be precise rather than absolute.

Regulators do not classify a whole company. They classify each function by what it does and what it claims to do. Tend has two functions that sit on opposite sides of the line.

The sensing layer can stay outside device regulation. Contactless monitoring of activity, sleep, and movement, framed as wellness and not tied to a named disease, falls under the low-risk general wellness policy.

The escalation function is the sensitive one. The moment software issues a clinical alert on its own, it looks like a medical device. Our design keeps a human, not the software, making that call.

It is that we design each function to sit on the safe side of the line, and where it cannot, we notify rather than seek approval.

United States: the wellness and decision-support line

The US Food and Drug Administration draws a clear boundary, and it is the boundary our product design respects.

What stays clear of regulation. Since the 21st Century Cures Act, software that maintains or encourages a healthy lifestyle and is unrelated to diagnosing or treating disease is excluded from the device definition.²¹

The FDA's 2026 general wellness guidance confirms that non-invasive sensors measuring activity, sleep, or pulse generally qualify as low-risk wellness products, as long as claims avoid disease references.²²

Where the line is crossed. The same guidance is explicit that a product is regulated if it provides diagnostic alerts, triggers clinical interventions, or recommends treatment.²³

Tend's escalation is exactly the kind of function that would cross this line if the software made the clinical decision. Our design keeps it on the safe side in two ways.

A licensed clinician, not the software, interprets the signal and decides to act. The software surfaces information; the human makes the call.

The escalation logic is transparent to that clinician. The FDA has said the more a tool is a black box to the professional, the more likely it is treated as a device.²⁴

This mirrors the FDA's clinical decision support carve-out, which keeps software outside the device definition when a professional can independently review the basis for a recommendation rather than relying on it blindly.²⁴

The honest caveat. This is a design discipline, not a loophole. If we ever marketed Tend as catching falls automatically, or let the software escalate without a clinician, the exemption would fall away. Our liability position and our regulatory position are the same discipline.

New Zealand: no pre-market approval, notification only

New Zealand is our beachhead, and its regime is materially lighter than the US or EU. This strengthens the case for launching here first.

There is no approval step. Under the Medicines Act 1981, there is no pre-market approval system for medical devices, and no requirement for any regulator to approve a device before it is supplied.²⁵

The only duty is notification. Medsafe requires that devices be notified to the WAND database within 30 days of a sponsor taking them to market. Notification is mandatory, but it is not an assessment of safety or performance.²⁵ ☐ ²⁶

The practical consequence is important. Even if part of Tend's stack were treated as a low-risk device in New Zealand, the obligation is to notify, not to win approval. That removes the long approval timelines that would otherwise gate a pilot.

We will still hold safety documentation, because Medsafe can request it, and we will design to that standard from the start.²⁶

European Union: the reason we minimise biometrics

The pre-read notes that competitors relying on facial recognition and emotion detection carry heavy EU exposure. The official basis for that is worth stating, because it is a deliberate design choice for us.

Emotion recognition is high-risk by default. The EU AI Act lists AI systems intended for emotion recognition among its high-risk category, which carries strict obligations.²⁷

There is a narrow exception for systems designed for medical or safety purposes, which a safety-focused eldercare use may fit, but high-risk obligations still apply.²⁸

Biometric data is doubly protected. Under the GDPR, biometric data used to identify a person is a special category with heightened protection, layered on top of the AI Act.²⁹

Our design minimises biometric processing. We favour contactless movement and vital-sign sensing over facial recognition and emotion inference. That keeps us clear of the heaviest EU obligations that our competitors invite.

The offshore clinician: an open question we flag honestly

The long-term model envisions a remote nurse, possibly offshore, triaging into many homes. This raises scope-of-practice and licensing questions that vary by country and are not yet settled.

A clinician who assesses and directs care into a patient's home is generally practising in the patient's jurisdiction, which usually requires local registration. Triage and escalation may be treated differently from hands-on care, but the boundary is untested for this model.

We do not claim this is solved. The offshore ratio that drives long-run margin depends on regulatory clarity we will pursue with counsel, not assume.

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